ClipSAT : EST II Math Level 1 [ANSWER KEY]

Unit 0.01: EST II Math Level 1 Diagnostic & Strategy Drill

Q1. Consider the function gx = 2x + 3 . What is the value of g4 ?

[ ] A: 7
[ ] B: 8
[x] C: 11
[ ] D: 14

Solution: To find the value of g4 , we substitute x = 4 into the function gx = 2x + 3 . g4 = 24 + 3 g4 = 8 + 3 g4 = 11 Therefore, the value of g4 is 11.

Q2. In the circle shown below, chords AB and CD intersect at point P. If PA = 4, PB = 6, and PC = 3, what is the length of segment overline{PD}?

[ ] A: 5
[ ] B: 7
[x] C: 8
[ ] D: 9

Solution: By the Intersecting Chords Theorem Power of a Point Theorem, when two chords AB and CD intersect at an interior point P, the products of their segments are equal: PA cdot PB = PC cdot PD Substitute the given values into the equation: 46 = 3PD 24 = 3 cdot PD PD = frac{24}{3} = 8 Common error analysis: - Adding the segment lengths instead of multiplying PA + PB = PC + PD implies 4 + 6 = 3 + PD implies PD = 7 yields 7, which fails because the segments satisfy a multiplicative relationship derived from similar triangles triangle PAD sim triangle PCB.

Q3. In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50^circ, what is the measure of angle GHD?

[ ] A: 50
[x] B: 130
[ ] C: 40
[ ] D: 90

Solution: Since line AB is parallel to line CD , angle AGH and angle CHG are alternate interior angles, so mangle CHG = mangle AGH = 50^circ . Since angle CHG and angle GHD form a linear pair along line CD , they are supplementary: mangle GHD = 180^circ - 50^circ = 130^circ.

Q4. The bar chart below shows the number of students who chose different sports as their favorite. What is the total number of students surveyed?

[ ] A: 50
[ ] B: 60
[x] C: 70
[ ] D: 80

Solution: To find the total number of students surveyed, we need to sum the number of students for each sport. From the bar chart: - Soccer: 20 students - Basketball: 15 students - Tennis: 10 students - Swimming: 25 students Total students = 20 + 15 + 10 + 25 = 70 Therefore, the total number of students surveyed is 70.

Q5. Consider the function gx = 2x + 3 . What is the value of g4 ?

[ ] A: 7
[ ] B: 8
[x] C: 11
[ ] D: 14

Solution: To find the value of g4 , we substitute x = 4 into the function gx = 2x + 3 . g4 = 24 + 3 g4 = 8 + 3 g4 = 11 Therefore, the value of g4 is 11.

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Q6. In the circle shown below, chords AB and CD intersect at point P. If PA = 4, PB = 6, and PC = 3, what is the length of segment overline{PD}?

[ ] A: 5
[ ] B: 7
[x] C: 8
[ ] D: 9

Solution: By the Intersecting Chords Theorem Power of a Point Theorem, when two chords AB and CD intersect at an interior point P, the products of their segments are equal: PA cdot PB = PC cdot PD Substitute the given values into the equation: 46 = 3PD 24 = 3 cdot PD PD = frac{24}{3} = 8 Common error analysis: - Adding the segment lengths instead of multiplying PA + PB = PC + PD implies 4 + 6 = 3 + PD implies PD = 7 yields 7, which fails because the segments satisfy a multiplicative relationship derived from similar triangles triangle PAD sim triangle PCB.

Q7. Given the function fx = 2x + 3 . What is the value of f^{-1}7 ?

[ ] A: 1
[x] B: 2
[ ] C: 3
[ ] D: 4

Solution: Let y = fx . Then y = 2x + 3 . To find the inverse function f^{-1}y , we swap x and y and solve for y . So, x = 2y + 3 . Subtract 3 from both sides: x - 3 = 2y . Divide by 2: y = frac{x - 3}{2} . Thus, f^{-1}x = frac{x - 3}{2} . We need to find f^{-1}7 . Substitute x = 7 into the inverse function: f^{-1}7 = frac{7 - 3}{2} = frac{4}{2} = 2 .

Q8. In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50^circ, what is the measure of angle GHD?

[ ] A: 50
[x] B: 130
[ ] C: 40
[ ] D: 90

Solution: Since line AB is parallel to line CD , angle AGH and angle CHG are alternate interior angles, so mangle CHG = mangle AGH = 50^circ . Since angle CHG and angle GHD form a linear pair along line CD , they are supplementary: mangle GHD = 180^circ - 50^circ = 130^circ.